## Questions

**Q1.**

**Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).**

This question is about static electricity and electric fields.

Figure 1 shows three charged objects, X, Y and Z.

**Figure 1**

Y has a positive charge and is held stationary.

The charge on Y causes X and Z to move in the directions of the arrows.

Which row of the table is correct for the charges on X and Z?

**(1)**

| | charge on X | charge on Z |
|---|---|---|
| ☐ **A** | negative | negative |
| ☐ **B** | negative | positive |
| ☐ **C** | positive | negative |
| ☐ **D** | positive | positive |

**(Total for question = 1 mark)**

**Q2.**

Figure 5 shows a metal chair being sprayed with paint.

The paint droplets come from a gun with an electric charge.

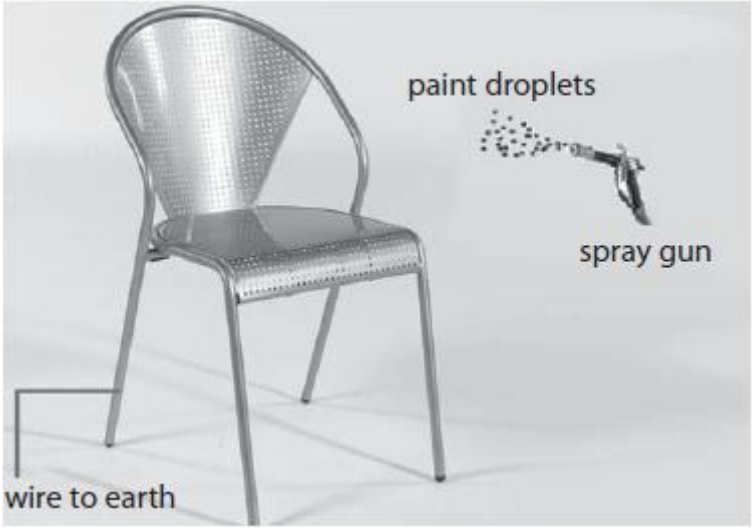


**Figure 5**

Inside the spray gun, electrons move along a charged wire towards the nozzle to charge the paint.

The charged paint droplets are sprayed from the nozzle.

The chair is connected to earth.

Which row of the table shows the correct combination of the charges as the charged paint droplets get near to the chair?

**(1)**

| | paint droplets | chair |
|---|---|---|
| ☐ **A** | negative | negative |
| ☐ **B** | negative | positive |
| ☐ **C** | positive | negative |
| ☐ **D** | positive | positive |

**(Total for question = 1 mark)**

**Q3.**

Figure 7 shows two metal spheres.

Metal sphere A is fixed to a table.
Metal sphere B can be moved.

Metal sphere B is placed at a short distance from metal sphere A.

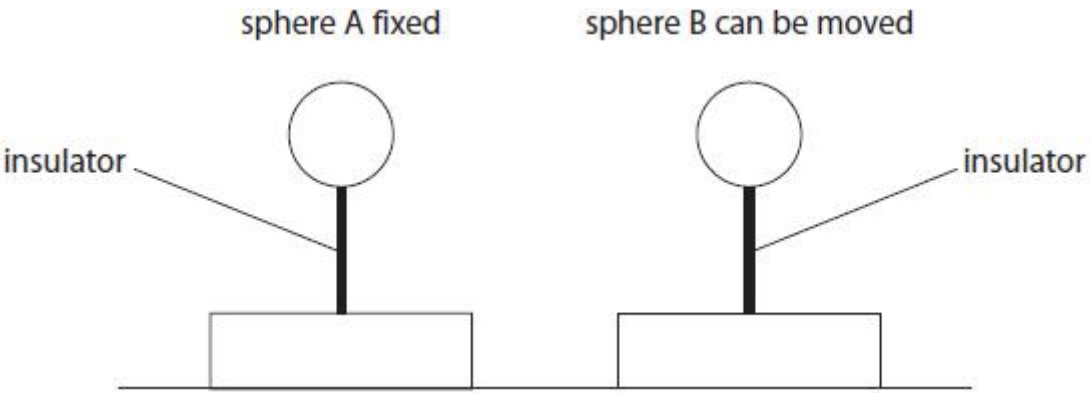


**Figure 7**

Both spheres are insulated from the table and given a negative charge.

The force between the charged spheres is measured.

(i) Explain, in terms of electric fields, why a force is exerted on sphere B.

**(2)**

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(ii) Sphere B is moved and the force between the spheres is measured at several different distances. Figure 8 is a graph of force on sphere B against distance between the centres of the spheres.

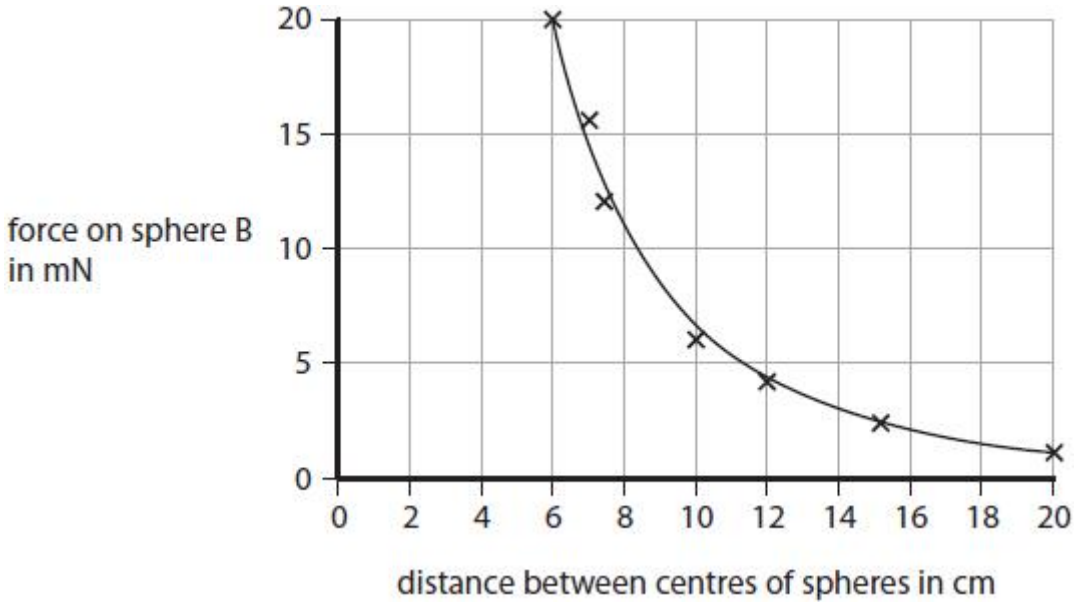


**Figure 8**

Describe how the force on sphere B varies with the distance between the centres of the spheres.

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**(Total for question = 4 marks)**

**Q4.**

**Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.**

This question is about static electricity.

Figure 9 shows a plastic block and a metal disc with an insulating handle.

The top surface of the plastic block has a negative charge.

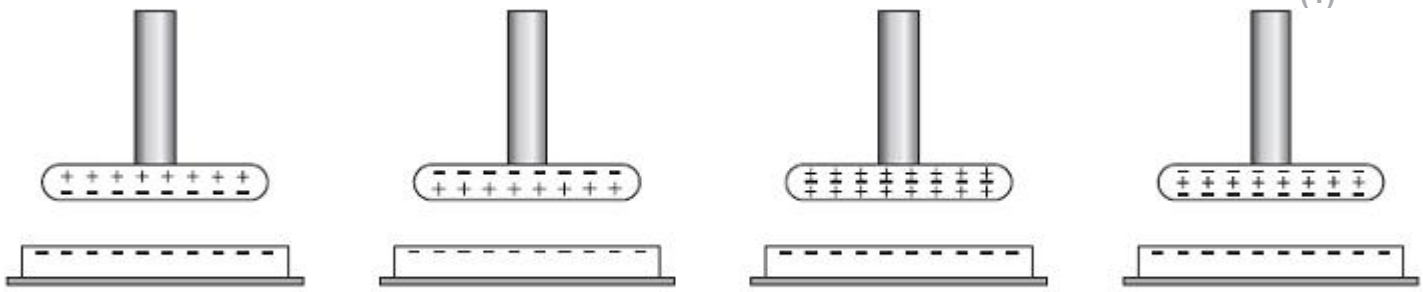
The metal disc has no overall electric charge.

**Figure 9**

A student uses the insulating handle to hold the metal disc above the plastic block.

(i) Which of these diagrams shows how the charge is distributed on the metal disc?

**(1)**



☐ **A** ☐ **B** ☐ **C** ☐ **D**

(ii) The student keeps holding the metal disc above the charged plastic block and taps the metal disc with a finger.

This earths the metal disc for a short time.
Explain why the disc now has an overall positive charge.

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(iii) Figure 10 shows the charges on part of the metal disc and the plastic block.

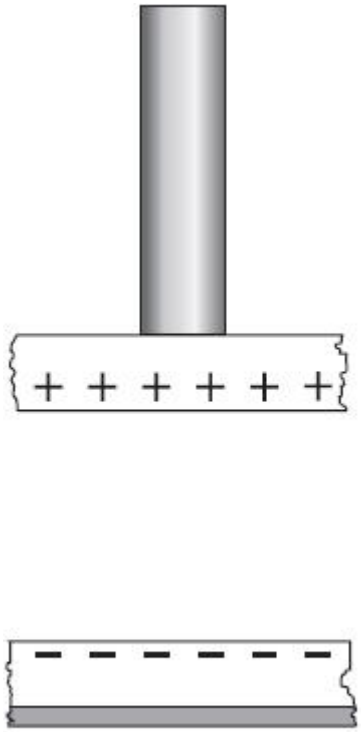

**Figure 10**

On Figure 10, draw lines to show the shape and direction of the electric field between the metal disc and the plastic block.

**(2)**

**(Total for question = 5 marks)**

**Q5.**

Figure 1 shows a prize that is made from a metal star on a plastic base.

**Figure 1**

A person starts to clean the prize by rubbing the plastic base with a dry cloth.

The plastic base becomes positively charged and the cloth becomes negatively charged.

The person then holds the metal star and rubs it with the charged cloth.

The cloth loses its charge.

Explain why the cloth loses its charge.

**(2)**

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**(Total for question = 2 marks)**

**Q6.**

A student rubs a plastic ruler against a woolly jumper.

The student tests the ruler and finds it has a positive charge.

Explain how the ruler becomes positively charged.

**(2)**

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**(Total for question = 2 marks)**

**Q7.**

Explain how an aircraft can become electrically charged as it flies through the air.

**(2)**

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**(Total for question = 2 marks)**

**Q8.**

Figure 2a shows another gold leaf electroscope that has been given an overall negative charge.

A student connects the metal cap of the charged electroscope to earth with a piece of wire as shown in Figure 2b.

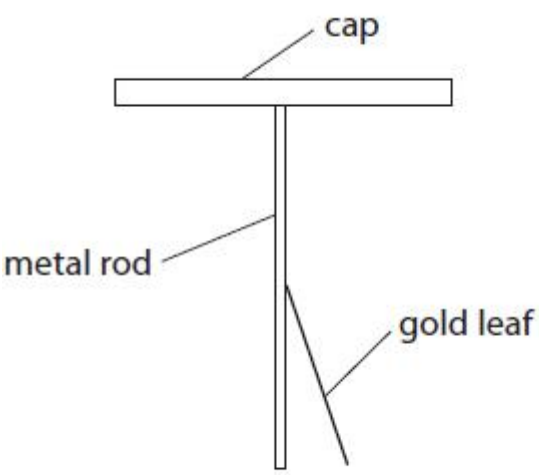


**Figure 2a**

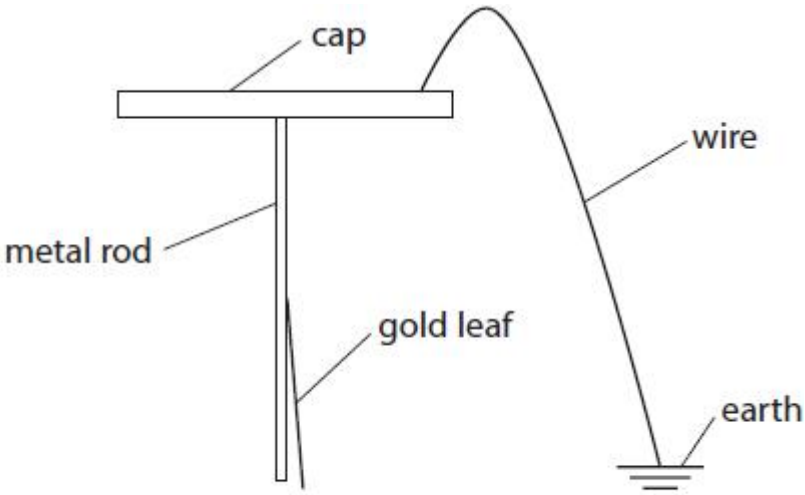


**Figure 2b**

Explain why the gold leaf has moved.

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**(Total for question = 2 marks)**

**Q9.**

An insecticide sprayer charges droplets of insecticide.

Figure 2 shows the sprayer being used to spray a leaf.

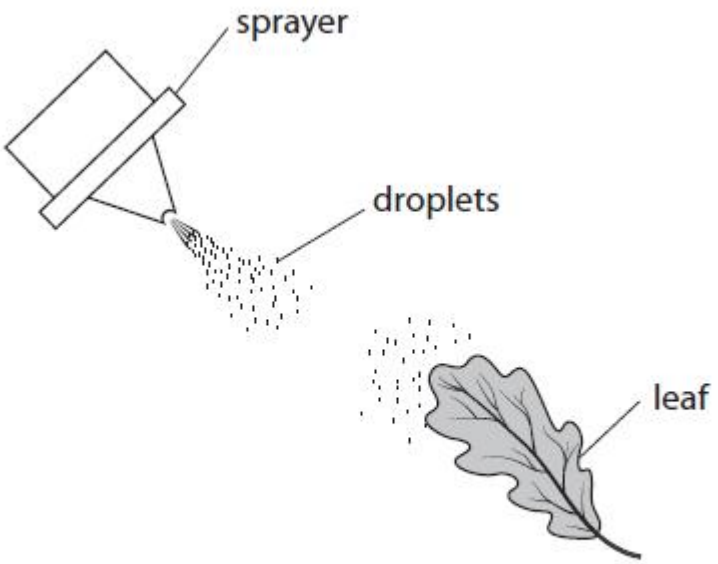


**Figure 2**

The leaf is connected to the ground (earthed).

Explain how charging the droplets helps to make sure that the leaf gets covered with insecticide.

You may add to Figure 2, including the sign (+ or –) of any charges, to help your answer.

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**(Total for question = 3 marks)**

**Q10.**

Figure 9 shows an electrostatic method for spray-painting a car door.

The car door has a negative charge.

The droplets of paint receive a positive charge as they leave the spray gun.

(Source: © Jens Brüggemann/123RF)

**Figure 9**

Explain how charging the door helps the paint to form an even coating on both sides of the door.

You should use ideas of forces and fields in your answer.

**(2)**

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**(Total for question = 2 marks)**

## Mark Scheme

Q1.

| Question Number | Answer | Mark |
|---|---|---|
| | **B** negative positive<br>The only correct answer is B correlating with attraction of X to Y, and repulsion of Z from Y. | **(1)**<br>**A01.1** |

**Q2.**

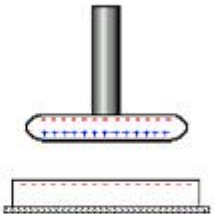
| Question number | Answer | Mark |
|---|---|---|
| | **B** negative positive | **(1)** |

**Q3.**

| Question Number: | Answer | Additional guidance | Mark |
|---|---|---|---|
| **(i)** | an explanation linking: | | **(2)**<br>AO 2 2 |
| | sphere A has an electric field (1) | both spheres have electric fields | |
| | sphere B is in it (1) | the electric fields interact/overlap<br><br>ignore nature of force; e.g. repulsion | |

| Question Number: | Answer | Additional Guidance | Mark |
|---|---|---|---|
| (ii) | a description to include:<br><br>as the distance increases the force (on the sphere B) decreases (1)<br><br>the greatest change is at smallest distances (1) | <br><br>negative correlation<br><br>non-linear<br>gradient changes<br><br>allow named non-linear functions such as exponential / inversely proportional in this context<br><br>reference to inverse square law scores 2 marks | (2)<br>AO 3 1a<br>AO 3 1b |

**Q4.**

| Question number | Answer | Additional guidance | Mark |
|---|---|---|---|
| (i) | B<br>  <br>A and D are incorrect because a negative charge cannot induce a negative charge<br>C is incorrect because the disc is insulated so negative charge cannot be removed | | (1) |

| Question number | Answer | Additional guidance | Mark |
|---|---|---|---|
| (ii) | an explanation linking:<br><br>electrons / negative charges move/ flow/transfer (1)<br><br>from the metal disc / to the student / to earth/ground (1) | reject positive charge moving for first mark<br><br>accept lose electrons | (2) |

| Question number | Answer | Additional guidance | Mark |
|---|---|---|---|
| (iii) | at least three straight lines joining disc and plastic (1)<br><br>arrow(s) from disc towards plastic (1) | judge by eye<br>ignore curved lines at edge<br><br>do not award mark if there are arrows in both directions | (2) |

**Q5.**

| Question number | Answer | Additional guidance | Mark |
|---|---|---|---|
| | An explanation linking **two** from<br><br>electrons transferred (from the cloth) to the metal (1)<br><br>metal is a conductor (1)<br><br>electrons travel through person / to earth (1) | accept negative charge for electrons<br><br>accept star for metal<br><br>electrons move through metal<br><br>(metal is ) earthed / grounded | (2)<br>**AO1** |

**Q6.**

| Question number | Answer | Additional guidance | Mark |
|---|---|---|---|
| | an explanation linking any **two** from<br><br>charged by **friction** (1)<br><br>electrons / negative charges rubbed off ruler (1)<br><br>woolly jumper becomes negative (1) | <br><br><br>electrons / negative charges transfer to jumper<br><br>ignore positive electrons / charges / particles | **(2)**<br>**AO2.1** |

**Q7.**

| Question number | Answer | Additional guidance | Mark |
|---|---|---|---|
| | An explanation that combines identification – understanding (1 mark) and reasoning/justification – understanding (1 mark):<br><br>• friction between aircraft and air (1)<br>• causes electron transfer between aircraft and air (1) | <br><br><br>accept idea of air rubbing against wings<br>ignore 'charge' and 'static'<br><br>do not allow (for second mark) idea of protons moving | **(2)** |

**Q8.**

| Question number | Answer | Additional guidance | Mark |
|---|---|---|---|
| | An explanation linking:<br><br>electrons move_(1)<br><br>with **one** from:<br><br>(leaf and/or rod) have been discharged (1)<br><br>(gold leaf) is no longer repelled (1) | <br><br>negative charges move<br>reject positive electrons for this mark<br><br><br>to the earth | **(2)**<br>**AO1** |

**Q9.**

| Question number | Answer | Additional guidance | Mark |
|---|---|---|---|
| | an explanation including any **three** from:<br><br>the leaf (becomes) charged (1)<br><br>opposite to charge on droplets / spray (1)<br><br>charges on leaf are **induced** (1)<br><br>opposite charges attract (1)<br><br>as a result of movement of electrons (between earth and leaf) (1)<br><br>idea of spray reaching parts of the back of the leaf (1)<br><br>charged droplets repel each other (having the same charge) (1) | **accept marks scored on diagram**<br><br>ignore polarity for this marking point<br><br>states charge opposite to what they have on the leaf<br><br><br><br><br>ignore positive electrons / charges / particles<br><br>allow all over | **(3)**<br><br>**AO1.1** |

**Q10.**

| Question number | Answer | Mark |
|---|---|---|
| | An explanation that combines identification – application of knowledge (1 mark) and reasoning/justification – application of understanding (1 mark):<br><br>• (negatively charged) door attracts (positively charged) paint (droplets) (1)<br><br>Plus any **one** of the following:<br><br>• therefore (positively charged) paint (droplets) follow lines of force and coat both sides of the car door (1)<br>• since electric field (or lines of force) directed towards the (car) door, then positive paint will move to the door (1)<br>• as electric field (or lines of force) touches all parts of the (car) door hence the positive paint will coat all parts of the door (1) | **(2)** |